Aircrack-ng

Introduction

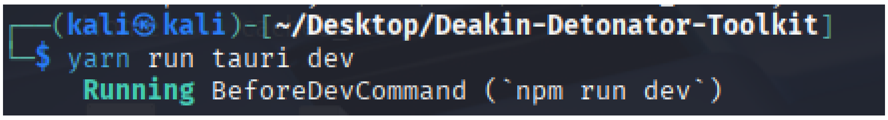
Aircrack-ng is a tool that analyses packet captures to attempt the recovery of WEP keys or WPA-PSK passphrases. Tools such as Airodump-ng or tcpdump capture the packets travelling between a wireless access point and another device on a wireless network encrypted with WEP or WPA-PSK. Aircrack-ng is deployed in the post capture phase of a wireless security assessment.

Disclaimer

Only run this tool against wireless networks that you own or on networks on which you have explicit written permission to perform security assessments. Intercepting network traffic and attempting unauthorised access to networks without explicit permission is unethical and illegal.

Setup and launch

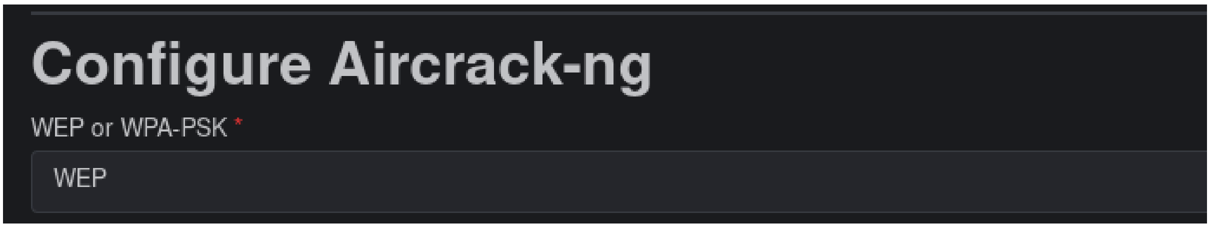
Initialise your virtual machine or device and open the terminal. Navigate to the directory where the Deakin Detonator Toolkit is installed. Then enter the following command into the terminal “yarn run tauri dev”.



Once the startup process has completed accept the user agreement and then navigate to the upper left hand of the Deakin Detonator Toolkit GUI and click on the tools button. Once in the tool selection screen select Aircrack-ng and proceed to the tool’s configuration panel.

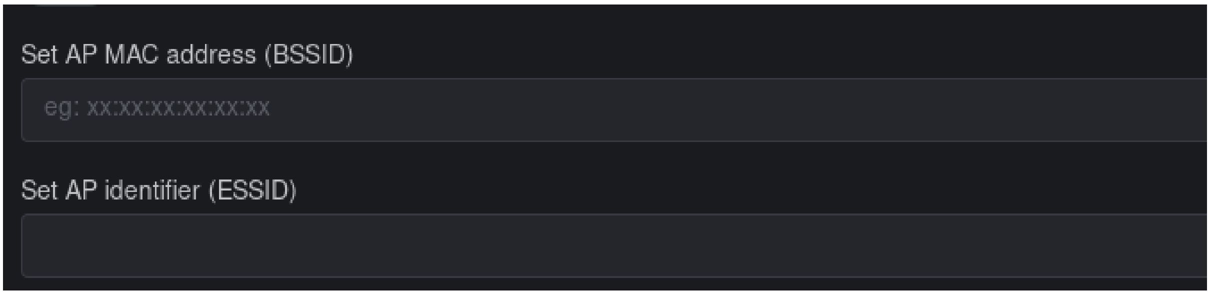
Configuration and Deployment

Once Aircrack-ng is launched select the encryption type you wish to crack, either WEP or WPA-PSK. This choice will be determined by the encryption type used to secure the network that was the target of the packet capture.

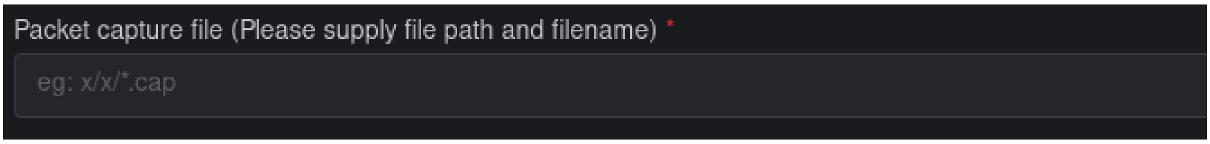


Once the encryption type is selected the BSSID also known as the MAC address of the target wireless access point can be entered and will accelerate the cracking process.

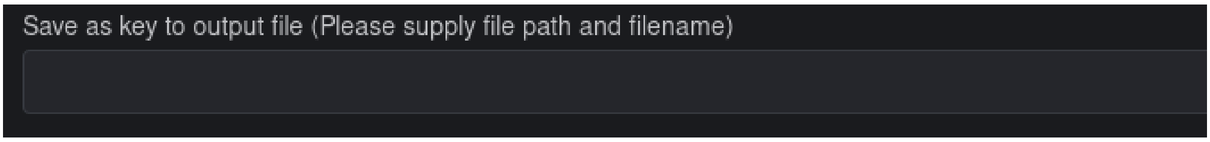
The BSSID is a unique twelve-digit number in hexadecimal format such as "0A:1B:2C:3D:4E:5F". MAC addresses are associated with the network interface card of devices capable of network access.



Once the above fields have been considered and filled out where necessary, enter the file path to the capture file that requires analysis. For example, Desktop/Captures/Ethical-Capture.cap. This step is required.



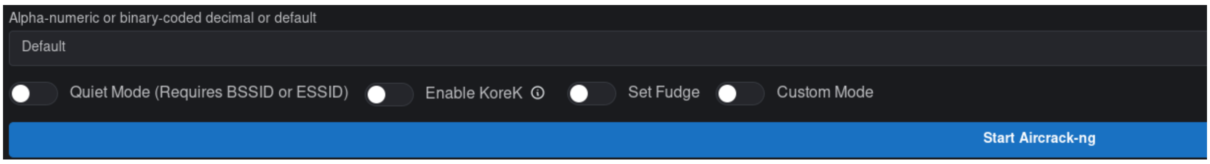
Optionally you may Input the location where the output of the cracking process is to be saved. For example, Desktop/DDT-Output/Cracked-Key.txt.



Finally, select the "Start Aircrack-ng" button to begin the process.

Advanced mode

When advanced mode is toggled on the following options become available.



Quiet mode is commonly used in high volume scanning operations as it removes the status and progress output to the terminal.

KoreK is an alternative statistical method used to analyse IVs and is recommended if the default PTW method fails to produce results as KoreK is generally slower than PTW.

Fudge enables an additional brute force function to the cracking process. Entering a number, the default is 2 extends the byte ranges considered when attempting to crack the key or passphrase.

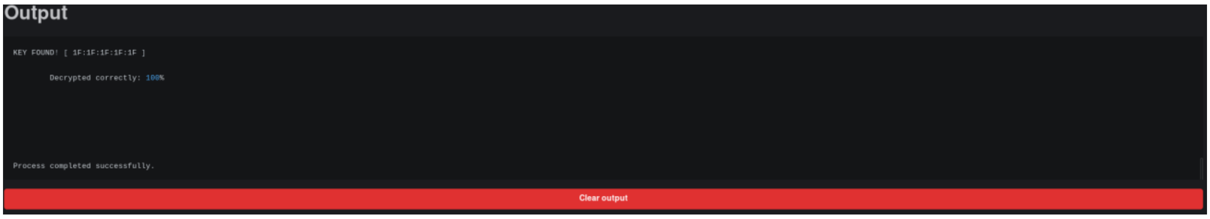
Alpha-numeric and binary encoded decimal differ from the default option which enables letters, numerals and symbols. Alpha-numeric only considers numerals and letters when cracking keys and passphrases. Binary encoded decimal only considers numerals when cracking keys and passphrases. Enabling the default option is recommended.

Custom mode allows you to append additional flags to the command line and utilise flags that are not available through the GUI.

Output

The output below is expected when a key or passphrase is successfully discovered. When a key is not found the output will be similar, simply stating "No key found".

To clear the output and start a new analysis click the clear output button and then select a new capture file to analyse, ensuring to clear the current fields manually or by closing Aircrack-ng and selecting the tool once again from the tool selection screen.



Troubleshooting

The most common issue encountered with Aircrack-ng is a poor-quality capture file. When cracking WEP, ensure that the capture file has the appropriate volume of IVs at least 40,000 but preferably 250,000 IVs to ensure the key is cracked. The more difficult the key the higher volume of IVs required to crack it.

When cracking WPA-PSK it is essential that the full four-way handshake is captured , if not then a new capture will be required.

When deploying a wordlist manually via the command line, ensure that the file path is correct as it is a common failure point that can be difficult to identify via standard troubleshooting.